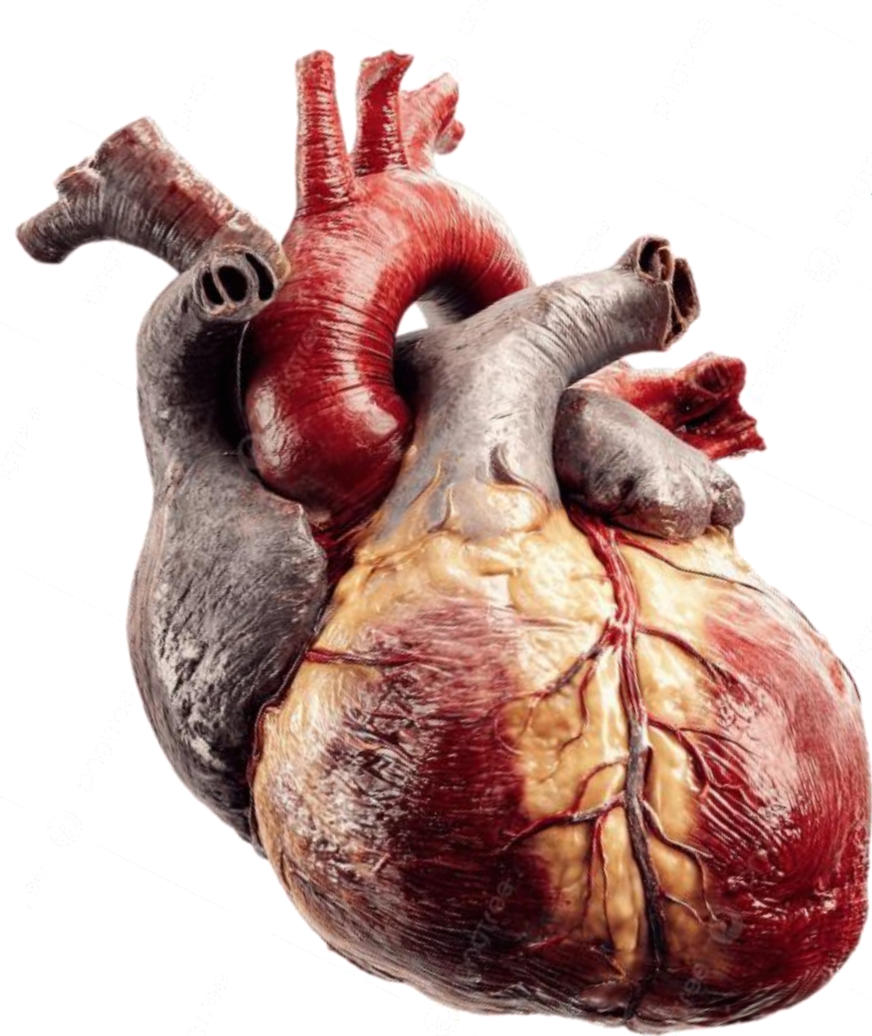




The Silent Burden of Heart Disease in Young America

A data-driven investigation into an overlooked condition where cardiomyopathy silently causes premature deaths among young people, demanding immediate attention.

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"The death of one person is a tragedy; the death of millions is a statistic."

— Joseph Stalin

Objectives

RESEARCH QUESTION

How does cardiomyopathy quietly drive mortality rates among America's youth compared to broader heart disease trends?

CORE OBJECTIVE

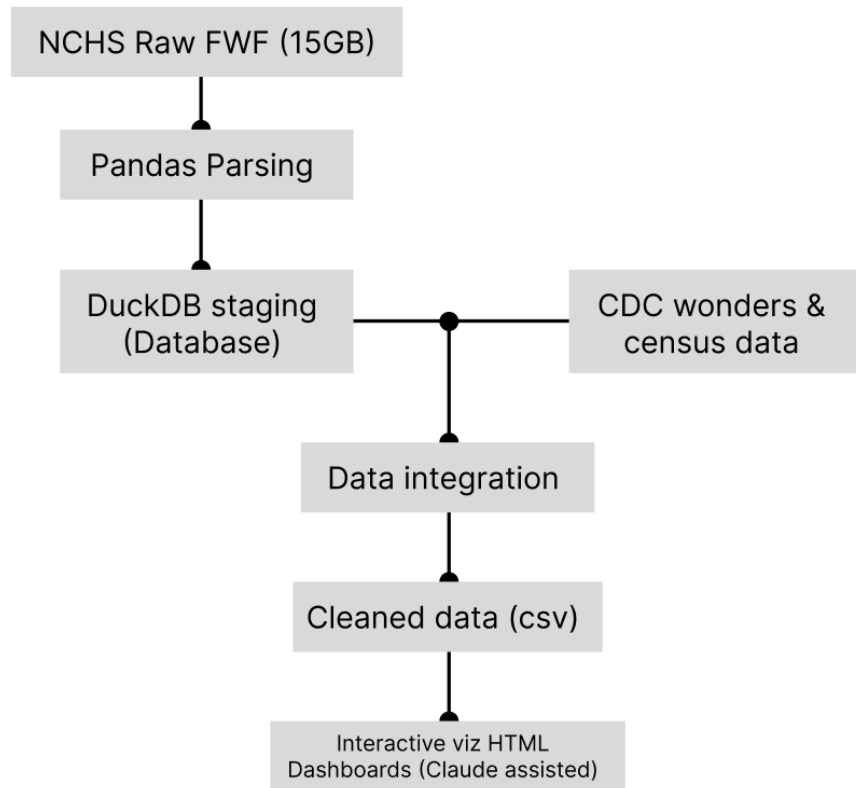
To challenge the prevailing assumption that heart disease is strictly an older adult's affliction.

Study Parameters

- Timeframe: 2018 - 2024
- Target age: 0 – 44
- Data Focus: The broad NCHS "Diseases of Heart" category with most common heart disease (Cardiomyopathy, Heart Failure, Chronic Ischemic heart disease, and heart attack) to subcategory Cardiomyopathy.
- Geography: US 50 states + Washington DC
- Sex: Male and Female (by birth)
- Race (experimental): Black, White, American Indian and Alaskan Native, Asian (Chinese, Japanese, Korean, Vietnamese, Filipino, Asian Indian, Other Asian), Pacific Islanders (Hawaiian, Samoan, Guamanian or Chamorro, Other Pacific Islander), Other / multi racial

Data Architecture & Methodology

Data Pipeline Architecture



Methodology

- Analyze longitudinal mortality trends across four major heart diseases, with a specific, deep-dive focus on Cardiomyopathy impacts within young populations (ages 0-44).
- Aggregated and parsed 22,043,274 total mortality records spanning a 7-year timeline (2018–2024).
- Tracked 4,774,283 deaths attributed broadly to heart disease (Cardiomyopathy, heart failure, chronic ischemic disease, and heart attack).
- Isolated 139,810 deaths driven specifically by Cardiomyopathy for targeted demographic and geospatial modeling.
- Detected and corrected a structural shift in the raw data starting in 2021 where the tape location for the "Race" column moved from positions 445-446 to 489-490.
- Translated coded values (ICD-10, demographics) into human-readable text to prepare the dataset for visualization.

Visualization & Deliverables Engine

Dashboard Output	Visualization Type	Key Insight & Focus
all_state_all_HeartDisease_all_age	Time-Series Heatmap	Geographic distribution of all 4 heart diseases across all 50 states + DC (2018-2024).
all_heart_disease_all_age	Stacked & Line Charts	Cardiomyopathy dominates younger populations (~60% of cases) but fades as a primary cause in older demographics.
i42_by_state_adults_only	Time-Series Heatmap	Geographic hot-spots isolating Cardiomyopathy deaths specifically in the 0-44 age bracket.
sex_and_seasonal_spike_chart_i42_adults_only	Temporal Line Charts	Male vs. Female death disparities and month-to-month (Jan-Dec) mortality spikes in young populations.

Why Rates, Not Just Counts

100 deaths in a city of 1,000 is a crisis. 100 deaths in a city of 1,000,000 is a footnote. Raw counts mislead. Mortality rates reveals true burden

$$\text{Mortality Rate per 100K} = \frac{\text{Deaths}}{\text{Population}} \times 100,000$$

Numerator

CDC NCHS microdata. Individual death records by ICD-10 code

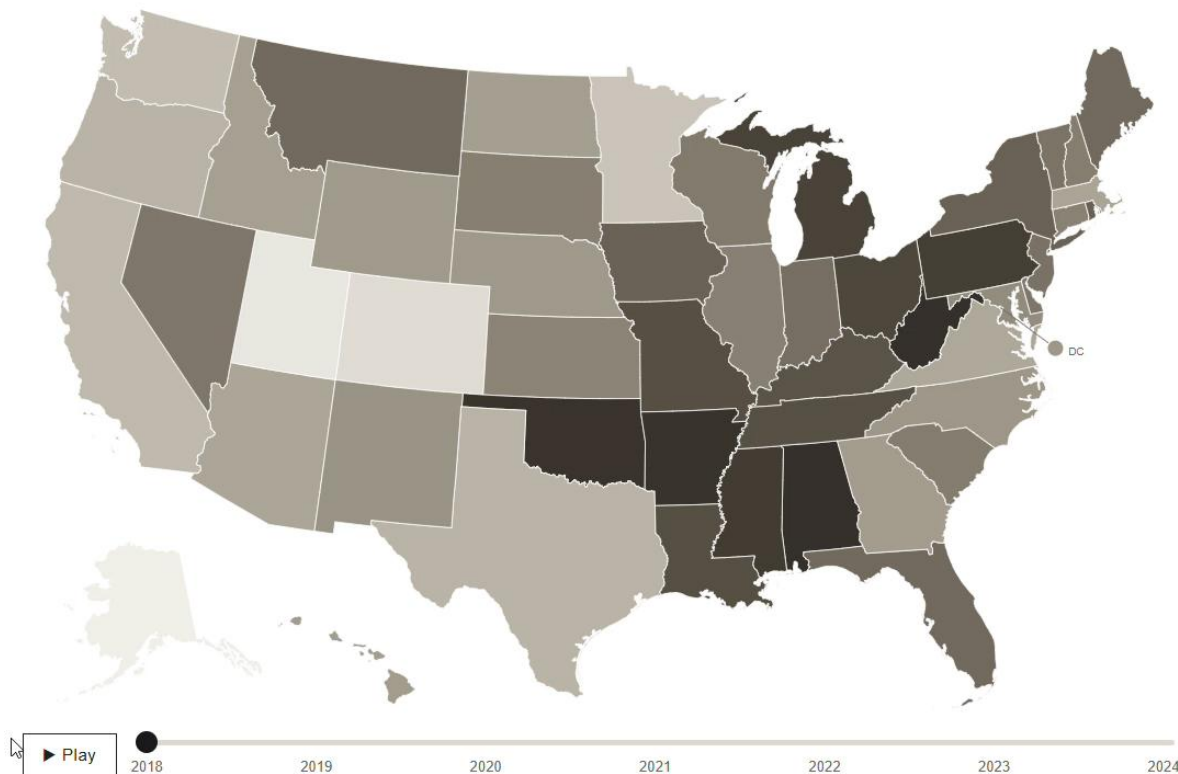
Denominator

U.S. Census Bureau population estimates by state, age, race, and sex

What the data reveals

Why this matters?

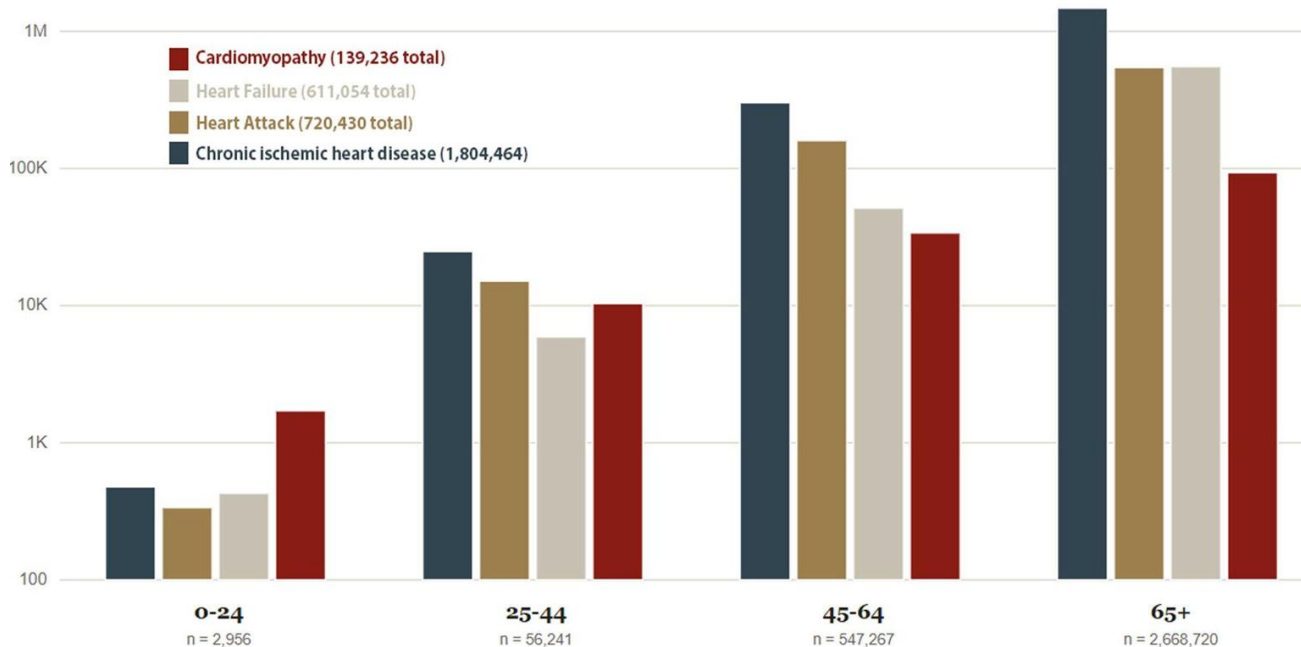
Heart Disease Affects Every Corner of the Nation



Includes death from the most common forms of heart diseases (Cardiomyopathy, Chronic Ischemic heart disease, Heart Failure and Heart Attack)

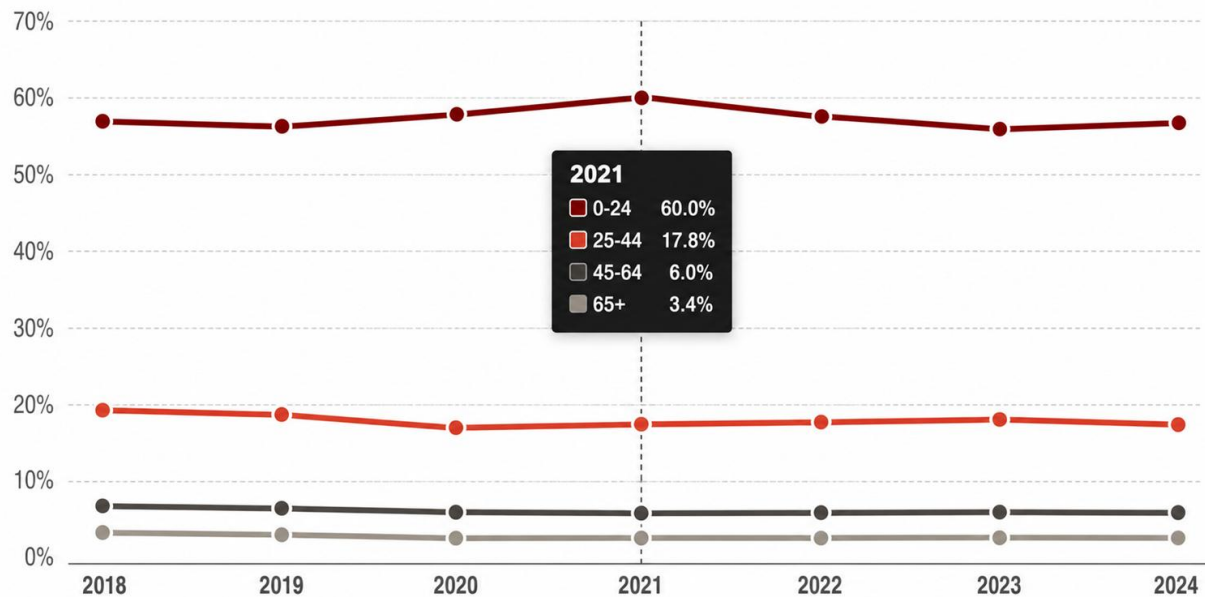
Data on aggregated scale:
National Average: 205.3 per 100k
Highest: West Virginia (294.6)
Lowest: Alaska (123.5)
Total death: 4,774,283

Heart Disease Doesn't Wait for Old Age



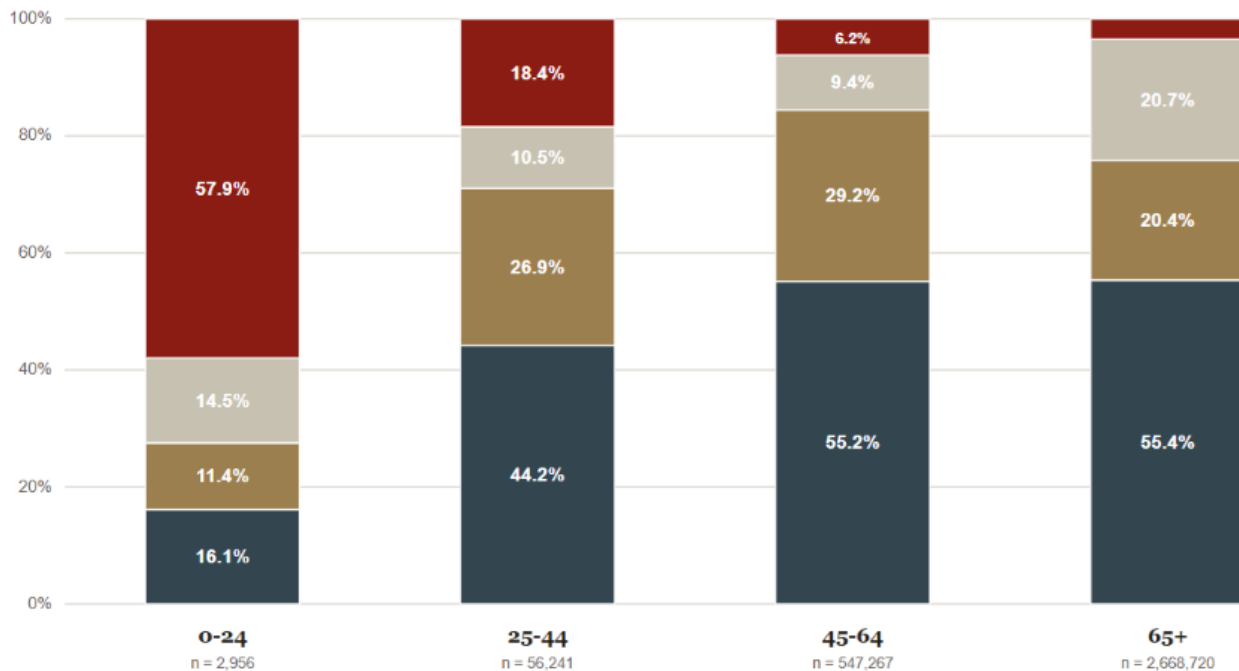
While most heart disease rise with age, Cardiomyopathy claims a large share of death in children and young adults before fading into older age groups

The pattern holds in every single year



Of 4.77M heart disease deaths, Cardiomyopathy dominates the 0-24 age groups with approx. 60% of death every single year

Youth Is Defined by Cardiomyopathy



Cardiomyopathy (139,236 total)
Heart Failure (611,054 total)
Heart Attack (720,430 total)
Chronic Ischemic heart disease (1,804,464)

Cardiomyopathy dominates the Youngest heart but its share is rapidly declined with age and ultimately replaced by Chronic Ischemic heart disease and Heart attack

A Silent Burden Across the Nation

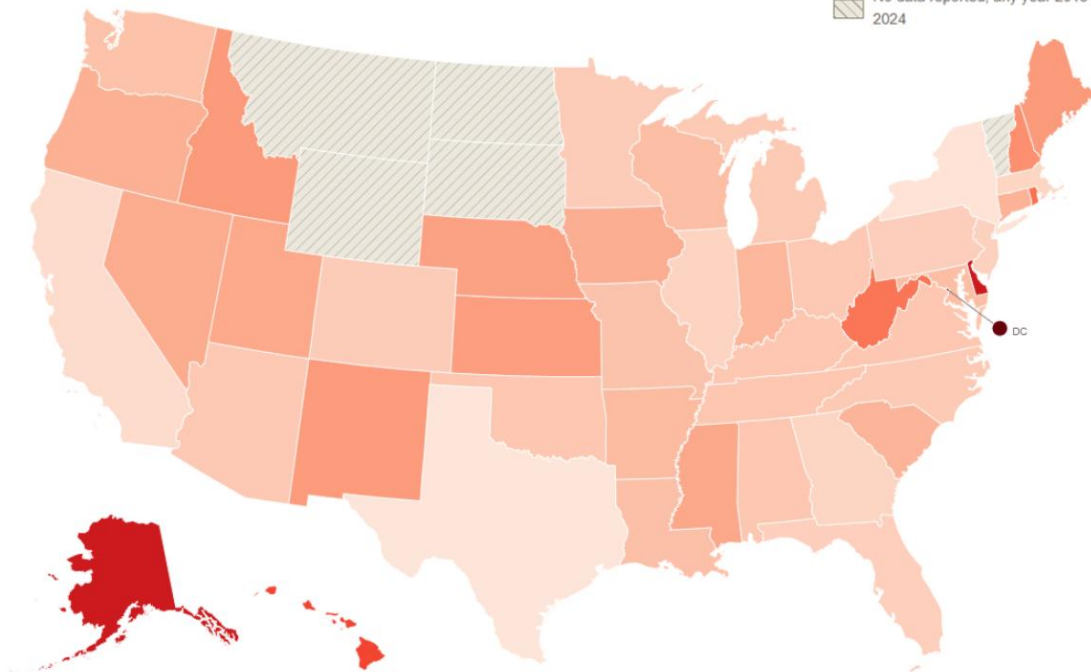
2018-2024 Aggregated

Deaths attributed to cardiomyopathy (ICD-10 code I42)

Rate per 100k

2.6 (lowest) 29.8 (highest)

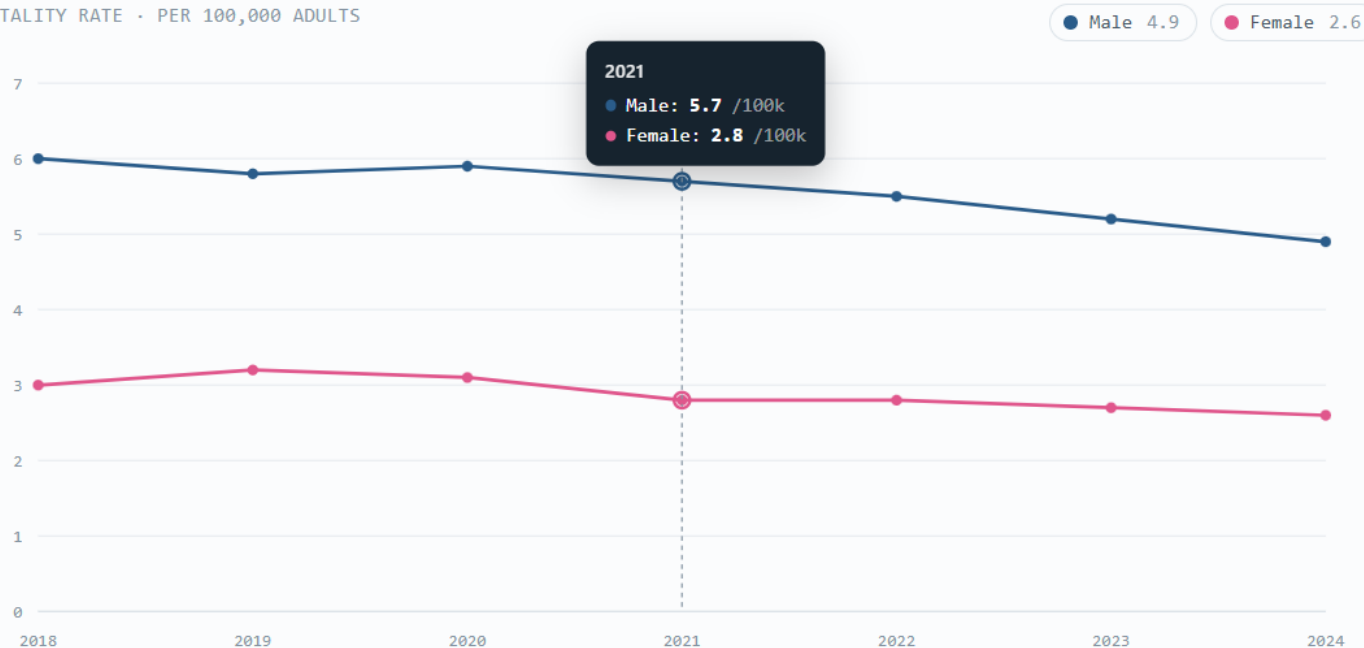
No data reported, any year 2018-2024



Cardiomyopathy is spread across the nation with national average of 4.5 per 100k. Washington DC with highest rate of 29.8 and Texas with lowest of 2.6.

Men Bear Double the Mortality Burden

MORTALITY RATE - PER 100,000 ADULTS

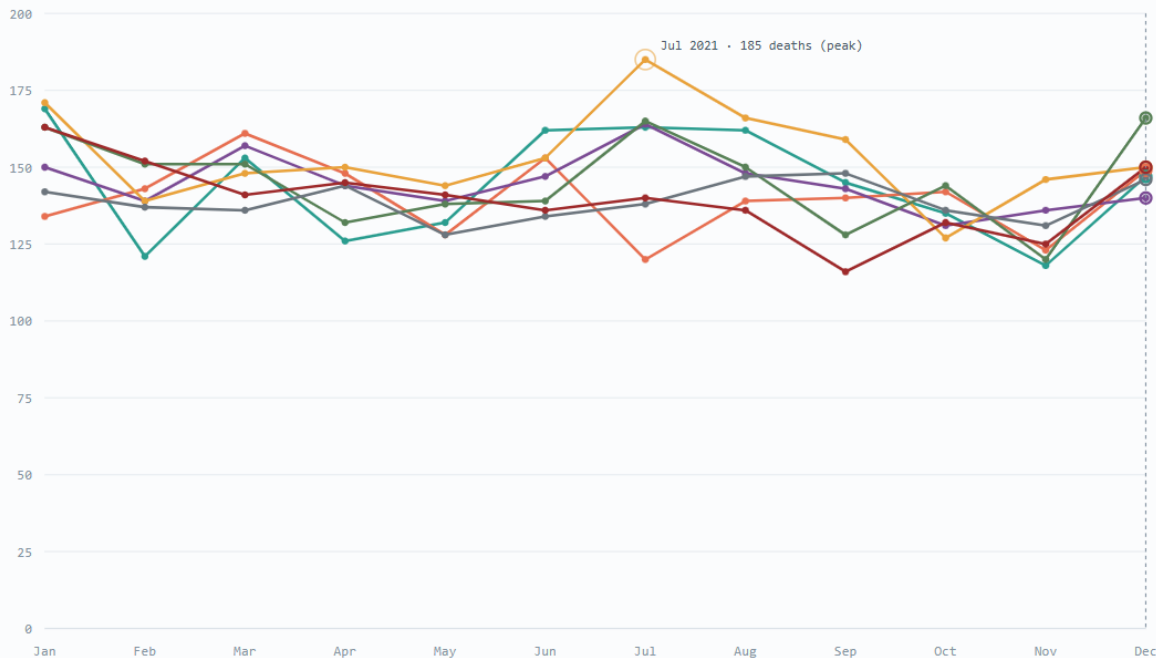


Approx 2.1x Avg
Male to Female
ratio

The male mortality rate fell from 6.4 to 5.3 per 100k (-17.2%) between 2018 and 2024. The female rate fell more modestly, from 3.0 to 2.6 per 100k (-13.3%), holding the gap between the two lines roughly constant across the period.

January is the peak month

DEATHS · BY MONTH, ONE LINE PER YEAR



December

2018: 147
2019: 149
2020: 140
2021: 150
2022: 166
2023: 146
2024: 150

Averaged across all seven years, January is the highest mortality month with sharp increase from November

The single largest monthly count in the dataset is July 2021, at 185 deaths

Conclusion

Cardiomyopathy causes just 3% of all heart disease death (139,236) of 4.77 Million) but approx. 60% of its death occurs in age group of 0-24 in every single year from 2018 – 2024. Even though it's not a large killer, it is silently becoming a young killer

Cardiomyopathy often affects young people and can develop before symptoms become apparent. Yet, because heart disease is commonly perceived as an older adult's condition, awareness and screening efforts often fail to reach the very population most vulnerable to this condition.

Since young adult men face a notably higher cardiomyopathy risk than any other group, targeted action matters most here. Symptoms often go unnoticed until the condition becomes serious, making targeted awareness among this population potentially the highest impact intervention indicated by these data.

Additional information

- NCHS Vital Statistics Online Data Portal (Mortality Multiple Cause Files) : [Data Access - Vital Statistics Online](#)
 - Variable mapping (2018-2024): [NVSS - Public Use Data File Documentation](#)
 - CDC Wonder: [Multiple Cause of Death, 2018-2024, Single Race Request Form](#)
 - Population by age & sex 2018: <https://data.census.gov/table/ACSST1Y2018.S0101?q=Age+and+sex&y=2018>
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 - Population by age & sex 2024: <https://data.census.gov/table/ACSST1Y2018.S0101?q=Age+and+sex&y=2024>
 - Broad Asian population 2018: <https://data.census.gov/table?q=B02015&y=2018>
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 - Broad Asian population 2024: <https://data.census.gov/table?q=B02015&y=2024>
 - Broad Native Hawaiian and Other Pacific Islander population 2018: <https://data.census.gov/table?q=B02016&y=2018>
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 - Broad Native Hawaiian and Other Pacific Islander population 2024: <https://data.census.gov/table?q=B02016&y=2024>
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